

Md. Taslim Reza

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Bibliography

I have served at the **Islamic University of Technology (IUT)**, a subsidiary organ of the OIC, as a Professor in the Department of Electrical and Electronic Engineering from January 2022 to January 2025. I currently hold the position of **Senior Consultant at Imperial Engineering and Consultancy (IEC)**. Additionally, I am providing consultancy services to **Asian University of Bangladesh (AUB)** for the establishment of laboratories in the EEE Department, as well as supporting the overall approval process of the department. Earlier, I joined IUT as an Assistant Professor on 12 September 2012. I have done my Bachelor of Science in Electrical and Electronic Engineering from IUT in 2005. In 2008, I obtained the Master of Science in Technology from **Tampere University of Technology, Finland** with distinction. My Master's thesis was fully funded by **NOKIA Finland**. In 2019, I have received my Ph.D. from the EEE Department of IUT. I have published three research papers on my Ph.D. topics on referred journals. I have also published a number of international and national peer reviewed journals and attended international conferences. I have developed my expertise in Medical Image Processing, Medical Instrumentation, Signal Processing application for Communication Engineering and Optical Communication. During my higher studies, I have received Bangladesh Government scholarship, OIC partial scholarship and prestigious Rectors award from Tampere University of Finland. I undertook additional responsibilities as Head of Planning, Procurement and Development at IUT, and Assistant Provost of the male hall and cafeteria. In addition, I contributed to the university's administrative functions by serving on various institutional committees. Alongside these roles, I was entrusted with the responsibility of overseeing the Biomedical Laboratory in the Department of Electrical and Electronic Engineering (EEE), IUT.

Areas of Expertise

Academic	Medical Image Processing, Tissue Modeling Elastography Imaging Medical Instrumentation Digital Signal Processing Electrical Machines Power Electronics
Administrative	Planning, Procurement and Development Laboratory Setup and Management Establishing New Department or Program Student Hall Management Medical Center Management Sub-Station Management Organizing International Fests and Programs

Administrative Involvements

2025 – Present	EEE Lab Development Consultant, Asian University of Bangladesh (AUB)
2025 – Present	Senior Consultant, Imperial Engineering and Consultancy (IEC)
2022 – 2024	Head, Planning, Procurement and Development, IUT
2020 – 2022	Assistant Provost, Male Hall, IUT
2018 – 2022	Member, Medical Center Management Team, IUT
2019 – 2023	Member Secretary, Internal Rules and Regulation (IRR) Review Committee, IUT
2016 – 2024	Laboratory in Charge, Biomedical Lab, EEE Department of IUT
2015	Short Course Coordinator for EEE Department
2013	Short Course Official for EEE Department
2021 – 2024	Chairman, Program of Action under OIC for IUT
2019 – 2022	Chairman, CDS Management Committee
2014 – 2024	Worked for more than 50 different committees including Organogram Committee review committee of IUT

Academic Qualifications

- 2013-2019 *Doctor of Philosophy (PhD)*, Islamic University of Technology (IUT), Gazipur, Bangladesh.
- Title: "Breast Elastography Modulus Reconstruction from Simultaneous Acquisition of Stress and RF Echo Signal"*
- Overview: A successful modulus reconstruction algorithm is proposed for Quasistatic Elastography and validated comparing with recent published data. At the same time simulated breast tissue model is proposed and validated with outcomes. Algorithm for selecting best frame from Elastography sequences is proposed and validated.*
- 2006-2008 *Master's in Biomedical Engineering*, Tampere University of Technology (TUT) Tampere, Finland.
- CGPA: 4.03 (Ranking first, got Rector's award for outstanding result)*
- Title: "Vignetting artifact reduction for Digital Camera Image"*
- Overview: A successful vignetting reduction algorithm is proposed and implemented. The proposed algorithm reduced the artifact by using minimum amount of memory space. The thesis work is funded by NOKIA, Finland.*
- 2002-2005 *Bachelor of Science (B.Sc.) in Electrical and Electronic Engineering*, IUT, Gazipur, Bangladesh.
- CGPA: 4.81 out of 5.00 (With distinction and OIC scholarship)*
- 2001 *Higher Secondary Certificate (HSC)*, Notredame College, Dhaka, Bangladesh.
- Total Marks: 843 (Avail Bangladesh Government Scholarship)*
- 1999 *Secondary School Certificate (SSC)*, Khulna Zilla School, Khulna, Bangladesh.
- Total Marks: 840*

Scholarships and Awards

- 2008 Rector's award for outstanding result in Master's Program.
- 2003 OIC Scholarship for 3 years during Bachelor Program.
- 2002 Bangladesh Government Scholarship in Higher Secondary level.

2023 Best Paper for “Structure-Enhanced Translation from PET to CT Modality with Paired GANs. In Proceedings of the 2023 6th International Conference on Machine Vision and Applications”, ICMVA 2023, Singapore, Singapore, March 10-12, 2023. ACM, 2023.

Academic Employment

2026 *Professor, Department of Electrical and Electronic Engineering, Asian University of Bangladesh (AUB), Bangladesh (Subject to UGC approval of the EEE Department).*

2022- 2025 *Professor, Department of Electrical and Electronic Engineering, Islamic University of Technology (IUT) Bordbazar, Gazipur, Bangladesh.*

2019- 2022 *Associate Professor, Department of Electrical and Electronic Engineering, Islamic University of Technology (IUT) Bordbazar, Gazipur, Bangladesh.*

2012- 2019 *Assistant Professor, Department of Electrical and Electronic Engineering, Islamic University of Technology (IUT) Bordbazar, Gazipur, Bangladesh.*

2009-2012 *Assistant Professor, Department of Electrical and Electronic Engineering, American International University-Bangladesh (AIUB) Dhaka, Bangladesh.*

2005-2006 *Lecturer, Department of Electrical and Electronic Engineering, Ahsanullah University of Science and Technology (AUST) Dhaka, Bangladesh.*

Research Employment

2009 *Researcher, Department of Signal Processing, TUT Tampere, Finland.*

2008 *Research Assistant, Department of Signal Processing, TUT Tampere, Finland.*

2007-2008 *Research Assistant, Department of Signal Processing, TUT Tampere, Finland.*

2007 *Research Assistant, Institute of Measurement and Information Technology, TUT Tampere, Finland.*

2007 *Trainee Research Scientist, Valtion Teknillinen Tutkimuskeskus (VTT) Tampere, Finland.*

Book

2012 *“Vignetting artifact reduction for Digital Camera Image: Efficient way of digital image artifacts reduction”. LAP LAMBERT Academic Publishing GmbH & Co. KG Heinrich-Böcking-Str.6-8, 66121, Saarbrücken,*

Publications (Journal)

- 2023 Md. Taslim Reza, Nchouwat Ndumgouo Ibrahim Moubarak, Mohammad Rakibul Islam, Md. Raihanul Haque Khan and Mirza Muntasir Nishat (2023). "Optimization of breast cancer ablation volume by ultrasonic pressure field characterization." Sensing and Bio-Sensing Research, Volume 39, Article 100553.
- 2023 Tasnim Ahmed, Ahnaf Munir, Sabbir Ahmed, Md. Borhan Hasan, Md. Taslim Reza and Md. Hasanul Kabir (2023). "Structure-enhanced translation from PET to CT modality with paired GANs." Proceedings of the 6th International Conference on Machine Vision and Applications, 2023.
- 2022 Mirza Muntasir Nishat, Fahim Faisal, Ishrak J. Ratul, Abdullah Al-Monsur, Asif M. Ar-Rafi, S. M. Nasrullah, Md. Taslim Reza and others (2022). "A comprehensive investigation of the performances of different machine learning classifiers with SMOTE-ENN oversampling technique and hyperparameter optimization for imbalanced data." Scientific Programming, 2022.
- 2022 Mohammad Rakibul Islam, A. N. M. Iftekher, Fateha Noor, Md. Rezaul Hoque Khan, Md. Taslim Reza and Mirza Muntasir Nishat (2022). "AZO-coated plasmonic PCF nanosensor for blood constituent detection in near-infrared and visible spectrum." Applied Physics A: Materials Science & Processing, Volume 128, Issue 1, Article 86.
- 2022 Mohammad Rakibul Islam, Fahim Yasir, Md. Rakib Hossain Antor, Md. Hasibur Turja, Abdullah Rahman, Md. Rezaul Hoque Khan and Md. Taslim Reza (2022). "An eye-shaped ultra-sensitive localized surface plasmon resonance-based biochemical sensor." Plasmonics, 2022.
- 2020 Gulam Mahfuz Chowdhury, Md. Mahedi Hasan, Asif Ahmed, Md. Wahid Tousif Rahman, Md. Taslim Reza. "Effective Strain Image Sequence Selection by Using Semi-Automated Image Processing Technique." GUB Journal of Science and Engineering, Volume-7, Issue-1.
- 2020 M. Rezaul Hoque Khan, M. Ashraful Hoque, M. Taslim Reza, M. Saifur Rahman (2020). "System Imperfections on the Overall Performance of MC- CDMA-Based Cognitive Radio Network (CRN) System." Journal of Circuits, Systems and Computers, Volume-30, Issue-4.
- 2020 M. Rezaul Hoque Khan, M. Ashraful Hoque, M. Taslim Reza, M. Saifur Rahman (2020). "Performance Analysis of MC-CDMA Based UWB System for Beyond 4G." International Journal of Mathematics and Computer Science, Volume-15, Issue-3.
- 2019 Md. Taslim Reza, Muhammad Rezaul Hoque Khan and Mohammad Rakibul Islam (2018). "Modulus reconstruction by simultaneous data response of surface stress and strain distribution." International Journal of Computer Science and Network Security, Volume-19, Issue-5.

- 2018 Md. Taslim Reza, Abu Hena Md. Mustafa Kamal, Muhammad Rezaul Hoque Khan and Mohammad Rakibul Islam (2018). *"Young's Modulus Distribution Prediction Analysis of Elasticity Imaging Technique Based on Location of Tumors."* International Journal of Computer Science and Network Security, Volume-17, Issue-8.
- 2017 Muhammad Rezaul Hoque Khan, Muhammad Fokhrul Islam, Golam Sarwar, Md. Taslim Reza and Md. Ashraful Hoque (2017). *"Carrier generation using a dual-frequency distributed feedback waveguide laser for phased array antenna (PAA)."* Journal of the European Optical Society- Rapid Publications, Volume-13, Article number-30.
- 2017 Muhammad Rezaul Hoque Khan, Muhammad Fokhrul Islam, Golam Sarwar, Md. Taslim Reza and Md. Ashraful Hoque (2017). *"Performance analysis of microwave carrier for Ku-band satellite signal."* International Journal of Computer Science and Network Security, Volume-17, Issue-4.
- 2016 Riyasat Ohib, Samin Yeasar Arnob, Riazul Arefin, Mohammad Muhtady Muhaisin, Tanzil Bin Hassan, MR Amin, Md. Taslim Reza (2016). *"ENF Based Grid Classification System: Identifying the Region of Origin of Digital Recordings."* IEEE Signal Processing Cup 2016.
- 2015 Mohammad Rakibul Islam, Rifad Hossain, Md. Ziaul Haque Bhuiyan, Tahmeed Ahmed Margoob, Md. Taslim Reza, Kazi Khairul Islam (2015) *"Arrhythmia Detection Technique using basic ECG Parameters."* International Journal of Computer Applications, Volume-119, Issue-10, Page-12-15.
- 2015 Md. Taslim Reza , Rashid-Al-Mukaddim, Fahim Ahmed Zaman, Nafiul Islam (2015) *"Automatic ROI Generation for Lesion Segmentation of Breast Ultrasound Image using Directional Gradient."*, International Journal of Electrical, Electronics and Data Communication, Volume-3, Issue-6, Page-57-60.
- 2013 K M Mohibul Kabir, Md. Taslim Reza, Md. Ashraful Hoque (2013) *"Analysis and Comparison of Cubic Quantum Dot Nanocrystals of Different Materials."* International Journal of Engineering Trends and Technology (IJETT), Volume-6, Issue-4, Page-198-203
- 2013 Naheed Hossain, Md. Taslim Reza, Adnan Quaium, Mohammad Rakibul Islam, Kazi Khairul Islam (2013). *"Instrumentation of Tissue Impedance Measurement System and the Detection of Cancer Tissue based on Capacitance Spectra of the Prostate."* International Journal of Scientific & Engineering Research, Volume-4, Issue-12, Page-730-733.
- 2013 Md. Rifat Rayhan, Faysal, Mohammad, Md. Taslim Reza (2013). *"Improvement of a Traffic System using Image and Video Processing."* IOSR Journal of VLSI and Signal Processing (IOSR-JVSP), Volume-2, Issue-3, Page-1-8.

- 2012 Md Taslim Reza, Rinku Basak Hoque, Md Ashraful (2012). *"Effects of Light on the Intraocular Pressure of Aqueous Humor."* American Academic & Scholarly Research Journal, Volume-4, Issue-4.
- 2012 Md. Taslim Reza, S.M.Ferdous, Md. Nayeemul Hasan, Md. Rokonuzzaman, Kazi Firoz Ahmed, A.Z.M.Shahriar Muttalib (2012). *"A Low Cost surface Electromyogram (sEMG) signal controlled Differential Drive based Automated Wheel Chair for Disabled."* International Journal of Scientific and Engineering Research, Volume-3, Issue-2, Page-46-51.

Publications (International Conference/National Conference)

- 2023 Tasnim Ahmed, Ahnaf Munir, Sabbir Ahmed, Md. Bakhtiar Hasan, Md. Taslim Reza, Md. Hasanul Kabir (2023). "Structure-Enhanced Translation from PET to CT Modality with Paired GANs. In Proceedings of the 2023 6th International Conference on Machine Vision and Applications", ICMVA 2023, Singapore, Singapore, March 10-12, 2023. ACM, 2023.
- 2023 Mahin Khan Mahadi, Samiur Rashid Abir, Al-Muzadded Moon, Mohammad Adnan, Md. An-Nur Ibne Khan, Mirza Muntasir Nishat, Md. Taslim Reza and others (2023). "Machine learning assisted decision support system for prediction of prostate cancer." Proceedings of the 20th International Conference on Electrical Engineering/Electronics, 2023.
- 2022 Sakawat Hossen Rakib, S. M. Masum, Atika Farhana, Md. Abdullah Islam, Md. Fokhrul Islam and Md. Taslim Reza (2022). "Design of a low-cost ultraviolet disinfection unit to minimize the cross-contamination of COVID-19 in transport." Proceedings of the International Conference on Advancement in Electrical and Electronic Engineering (ICAEEE), 2022.
- 2021 Ahnaf Akif Rahman, Fahim Faisal, Mirza Muntasir Nishat, Md. Imran Siraji, L. I. Khalid, Md. Raihanul Haque Khan and Md. Taslim Reza (2021). "Detection of epileptic seizure from EEG signal data by employing machine learning algorithms with hyperparameter optimization." Proceedings of the 4th International Conference on Bio-Engineering for Smart Technologies (BioSMART), 2021.
- 2020 Sakawat Hossen Rakib, Md. Taslim Reza, Md. Fokhrul Islam (5-7 June 2020). *"Design of microstrip patch sensor for non-invasive body electrolyte monitoring."* Full Virtual Conference IEEE TENSYP 2020.
- 2018 Riyasat Ohib, Samin Yeasar Arnob, Riazul Arefin, Mohammad Muhtady Muhaisin, Tanzil Bin Hassan, MR Amin, Md. Taslim Reza (13-14 November 2018). *"ENF Based Grid Classification System: Identifying the Region of Origin of Digital Recordings."* International Conference on Electrical Engineering & Computer Science (ICEECS) 2018, Bali, Indonesia.

- 2015 Md. Taslim Reza, Rashid-Al-Mukaddim, Fahim Ahmed Zaman, Nafiul Islam (11-13 April, 2015). *"Automatic ROI Generation for Lesion Segmentation of Breast Ultrasound Image using Directional Gradient."* Twelfth International Conference on Technological Advances in Electrical, Electronics and Computer Engineering (ICTAEECE), Paris, France.
- 2011 S.M.Ferdous, Taslim Reza, Md. Nayeemul Hasan, Ahmed Al Mansur, Kazi Firoz Ahmed (21-23 October, 2011). *"A surface electromyogram (sEMG) signal guided control system of an automated wheel chair for disable using neck muscle movements."* Thirteenth international conference on electrical machines drives and power system, Varna, Bulgaria.

Affiliations

Member, Institute of Electrical and Electronic Engineers (IEEE) Member,
The Institution of Engineers, Bangladesh (IEB)
Life Time Member, IUT Alumni Association
Member, Club Notredamians

Languages

Bengali: Mother language
English: Fluent in Reading, Writing, Speaking and Listening

Date of Birth

1st November 1983

Referees

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